the volume and is the density. With 2 different liquids mass of equal volumes are proportional to their densities. If a volume V of one liquid produces n1 drops and other liquid presents n2 drops, then

2Π ry1n1=V ρ1and 2 Πr y2n2=Vρ2
Y2= (n1-n2) (ρ1-ρ2) y1

Y1 and Y2 are surface tensions of saturated liquid and liquid under experiment.
Procedure:-
The stalagmoneter is cleaned and dried. It is then filed by immersing the drop tip in water and sucking under water will rise above the mark. Then the screw pinch cork is adjusted till the rate of fall of drop is10-15 drops per minute from the end of the capillary tube. The stalagmometer is then refilled and the number of drops is found when the meniscus passes the upper mark until it just passes lower mark.
The stalagmometer is and dried and is filled with given liquids. The numbers of drops are found out as before. Density of the liquid is noted. Surface tension is calculated from the above equation.
Weight of 10 drops of water=.....
Weight of 1 drop of water=.....
Weight of 10 drops of the given liquid=.....
Weight of 1 drop=.....
Surface tension of water=.....
Surface tension of the given liquid=.....

Result:-Surface tension of the given liquid=.....
can you pls do that for thsi tooo

I'll simplify all four experiments for your lab exam. Here they are:

Experiment No. 4: Transition Temperature & Cryoscopic Constant

Principle (Simple)

- **Transition temperature** = temperature where a substance changes phase (solid to liquid), and it stays constant during the change.
- Adding a solute (dissolved substance) **lowers** this transition temperature.
- **Cryoscopic constant (K_f)** measures how much the temperature drops and is calculated using:

$$K_f = \frac{M \times W_1 \times \Delta T}{1000 \times W_2}$$

Where: M = molar mass of solute, ΔT = temperature drop, W_1 = solvent weight, W_2 = solute weight.

Procedure (Easy Steps)

1. **First Run (Pure salt):**
   - Take 10-15 g sodium thiosulphate crystals in a tube with thermometer and stirrer.
   - Heat in a beaker of hot water until it melts and temperature rises 5°C more.
   - Remove tube and record temperature every 30 seconds.
   - When temperature drops to 40°C, add one crystal and stir.
   - Temperature suddenly rises and stays constant—this is the **transition temperature** (note it).
   - Plot a time vs. temperature graph.
2. **Second Run (With solute):**
   - Add the given solute to the salt hydrate.
   - Repeat the same steps.
   - The transition temperature will be **lower** this time.
3. **Calculate:**
   - ΔT = difference between the two transition temperatures.
   - Use the K_f formula to get the cryoscopic constant.

Experiment No. 5: Acid-Base Titration by Conductance

Principle (Simple)

- **Conductance** = how easily a solution conducts electricity.
- HCl conducts well due to H^+ and Cl^- ions.
- When NaOH is added, conductance **decreases** until the acid and base neutralize (equivalence point).
- After that, adding more NaOH increases conductance due to extra OH^- ions.
- A graph of conductance vs. NaOH volume shows a dip—the lowest point is the equivalence point.

Procedure (Easy Steps)

1. **Setup (Find Cell Constant):**
   - Prepare 0.1 N KCl by dissolving 0.7456 g KCl in 100 ml distilled water.
   - Dilute 10 ml of 0.1 N KCl to 100 ml to make 0.01 N KCl.
   - Measure conductance of water, 0.1 N KCl, and 0.01 N KCl.
   - Calculate cell constant = specific conductance ÷ measured conductance.
2. **Main Titration:**

- Pipette 20 ml of 0.1 N HCl into a beaker and measure conductance.
- Add 2 ml of 0.1 N NaOH and measure conductance.
- Keep adding NaOH in 2 ml steps (4 ml, 6 ml, 8 ml, etc.) and measure each time.
- Plot a graph: conductance (y-axis) vs. volume of NaOH (x-axis).

3. **Find Result:**
   - The lowest point on the graph = equivalence point.
   - Read the volume of NaOH at this point.
   - Calculate the normality of NaOH if needed.

Experiment No. 6: Viscosity Measurement

Principle (Simple)

- **Viscosity** = how thick/sticky a liquid is (internal friction that resists flow).
- Using an **Ostwald viscometer**, we compare the flow time of different liquids.
- The relationship is:

$$\frac{\eta_1}{\eta_2} = \frac{\rho_1 \times t_1}{\rho_2 \times t_2}$$

Where: η = viscosity, ρ = density, t = time of flow.

Procedure (Easy Steps)

1. **Setup:**
   - Clean the Ostwald viscometer with acetone and dry it.
   - Pipette 10 ml of water into it.
   - Draw water into the upper bulb until the level matches the mark.
2. **Measure Flow Time:**
   - Let the water flow down, start a stopwatch, and note the time taken.
   - Repeat for other test liquids (ethanol, acetone, etc.).
3. **Measure Density:**
   - Find the density of each liquid.
4. **Calculate:**
   - Use the formula above to find the viscosity of the test liquids (if water's viscosity is known).

Experiment No. 7: Surface Tension

Principle (Simple)

- **Surface tension** = force that holds the liquid surface together (like a thin elastic sheet).
- When a drop falls from a dropper, surface tension holds it until it becomes too heavy.
- By counting the number of drops, we can calculate surface tension.
- The formula is:

$$\gamma_2 = \frac{(n_1 - n_2) \times (\rho_1 - \rho_2)}{1} \times \gamma_1$$

Where: γ = surface tension, n = number of drops, ρ = density.

Procedure (Easy Steps)

1. **Setup:**
   - Clean and dry the stalagmometer (dropper apparatus).
   - Fill it with water by sucking upwards past the mark.
   - Adjust the screw pinch cork so drops fall at 10-15 drops per minute.
2. **Count Drops of Water:**
   - Refill and count how many drops fall from the upper mark to the lower mark.
   - Weigh 10 drops of water.
3. **Repeat with Test Liquid:**
   - Fill the stalagmometer with the given test liquid.
   - Count the number of drops the same way.
   - Weigh 10 drops of the test liquid.
   - Find the density of the test liquid.
4. **Calculate:**
   - Use the formula to calculate the surface tension of the test liquid.
   - (Water's surface tension ≈ 0.073 N/m, you'll likely use this as reference.)

Good luck with your exam!